No uniform vanishing last iterate bound for exact Polyak steps

Abstract

For every prescribed number of updates, we construct a convex continuously differentiable function of two variables and an initial point at distance one from its unique minimizer. Both the function and its gradient have global Lipschitz constants at most one. Ordinary Polyak steps, using the exact optimal value and the actual squared gradient norm, produce a terminal function gap greater than $1/40$. Thus no class-uniform last-iterate bound tending to zero holds under these assumptions. The construction prescribes an exact backward itinerary for a quadratic ratio map and uses a conjugate extension to bound the gradient globally. Each run uses a single fixed objective; the objective and initialization depend on the horizon. We compare the result with Theorem 2.3 of the public preprint arXiv:2407.15195v1, identify the incompatible equal-gradient-norm reduction, and give an explicit finite-horizon contradiction to its displayed bound. The result is consistent with fixed-instance convergence and best-iterate guarantees.

Keywords: Polyak step size; last iterate; smooth convex optimization; worst case bounds

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1 The uniform terminal criterion

Let $\mathcal{C}$ be the class of convex functions $f : \mathbb{R}^2 \to \mathbb{R}$ that are continuously differentiable, satisfy

$$\|\nabla f(x) - \nabla f(y)\| \le \|x - y\|, \qquad |f(x) - f(y)| \le \|x - y\|, \tag{1}$$

for all $x, y \in \mathbb{R}^2$, and have unique minimizer 0 with $f(0) = 0$. All norms are Euclidean. We use the ordinary exact Polyak update

$$x_{k+1} = x_k - \frac{f(x_k)}{\|\nabla f(x_k)\|^2} \nabla f(x_k). \tag{2}$$

If the gradient is zero, the iterate is kept fixed. In particular the method knows the correct optimal value; no error, momentum, step cap or projection is present.

For $N \ge 1$, define

$$W_N = \sup\{f(x_N) : f \in \mathcal{C},\ \|x_0\| = 1,\ (x_k) \text{ follows } (2)\}. \tag{3}$$

The supremum allows the objective and initial point to depend on N. The objective remains unchanged during all N updates of each instance.

Theorem 1. *For every integer $N \geq 1$ there is an explicit instance in* (3) *for which*

$$f(x_N) \geq \frac{14010049}{514560000} > \frac{1}{40}. \tag{4}$$

Consequently $W_N > 1/40$ for every $N \geq 1$, so no upper bound depending only on N and tending to zero can hold for this class.

The dimension, smoothness constant, global function Lipschitz bound and initial radius are all fixed. No positive strong-convexity modulus is fixed across the family. The theorem concerns exact arithmetic and makes no claim about the robustness of its initialization to rounding.

Nonmonotonicity of Polyak steps in two dimensions is already known: Bauschke et al. [1, Example 9.4] give a nonsmooth example using distances to lines. The contribution here is the absence of a vanishing bound uniform in the horizon under the simultaneous global smoothness and Lipschitz assumptions in (1). Best-iterate guarantees for Polyak steps are classical; see, for example, Hazan and Kakade [2]. He et al. [3, Theorems 3.1–3.4 and Section 3.2] give tight best-iterate results and analyze special quadratic trajectories and their instability. We retain that ancestry. Our construction uses a prescribed backward itinerary to place a large function value at an arbitrary terminal time, with a global bounded-gradient extension. Orabona and D'Orazio [4, Section 6] give negative results involving different surrogate minima and discuss fixed-instance convergence. The distinctions between a terminal criterion, a best-iterate criterion and a single infinite orbit matter here.

The claimed last-iterate upper bound in Zamani and Glineur [5, Theorem 2.3] applies to the assumptions in (3). Section 4 compares the precise hypotheses and indexing and proves a finite contradiction. The comparison concerns the explicitly identified public preprint version, rather than a general claim about all Polyak variants or other results of its authors.

2 A globally smooth function with a quadratic core

Fix $0 < \mu \leq 1$ and write

$$q_\mu(x) = \tfrac{1}{2}(x_1^2 + \mu x_2^2), \qquad F_\mu(x) = \max_{\|p\| \leq 1} \left\{ \langle p, x \rangle - \tfrac{1}{2}(p_1^2 + p_2^2/\mu) \right\}. \tag{5}$$

The norm constraint bounds the global gradient without changing the relevant quadratic orbit.

Lemma 1. *The function F_μ belongs to $\mathcal{C}$. On the closed unit ball,*

$$F_\mu(x) = q_\mu(x), \qquad \nabla F_\mu(x) = (x_1, \mu x_2). \tag{6}$$

Every exact Polyak orbit of F_μ starting in that ball remains there.

Proof. Let $D = \mathrm{diag}(1, \mu^{-1})$. The maximum in (5) exists and has a unique maximizer p_x, since its objective is strictly concave in p. Convexity of F_μ follows from its representation as a supremum of affine functions. The variational inequalities for p_x and p_y give

$$\langle x - y, p_x - p_y \rangle \geq \langle D(p_x - p_y), p_x - p_y \rangle \geq \|p_x - p_y\|^2. \tag{7}$$

Thus $\|p_x - p_y\| \leq \|x - y\|$. Optimality also gives

$$0 \leq F_\mu(y) - F_\mu(x) - \langle p_x, y - x \rangle \leq \langle p_y - p_x, y - x \rangle \leq \|y - x\|^2. \tag{8}$$

Therefore F_μ is continuously differentiable with gradient p_x, and this gradient is globally 1-Lipschitz. Comparing the two maxima in (5), using $\|p\| \leq 1$, proves that the function itself is 1-Lipschitz.

Clearly $F_\mu(0) = 0$. If $x \neq 0$, a sufficiently small positive multiple $p = \varepsilon x$ is feasible and gives the positive value $\varepsilon\|x\|^2 - \varepsilon^2\langle Dx, x\rangle/2$. Hence the minimizer is unique. If $\|x\| \leq 1$, the unconstrained maximizer $D^{-1}x = (x_1, \mu x_2)$ is feasible. This proves (6), including the boundary.

For any convex differentiable f with $f(0) = \min f = 0$, convexity gives $\langle \nabla f(x), x \rangle \geq f(x)$. At a nonzero gradient, direct expansion of (2) yields

$$\|x_{k+1}\|^2 \leq \|x_k\|^2 - \frac{f(x_k)^2}{\|\nabla f(x_k)\|^2}. \tag{9}$$

At a zero gradient the convention also preserves the norm. Applying this inequality to F_μ proves unit-ball invariance before any use of the quadratic orbit. ∎

3 An exact itinerary for an arbitrary horizon

Fix an integer $N \geq 1$ and put

$$\mu = \frac{1}{100(N+1)}, \qquad H_\mu(z) = \frac{z(1 - 2\mu - z^2)}{(2-\mu)z^2 + \mu}. \tag{10}$$

Set $z_{N-1} = \sqrt{\mu}$. For $k = N-2, \ldots, 0$, choose z_k as the largest root in $[1/3, \sqrt{1-2\mu}]$ of

$$H_\mu(z_k) = z_{k+1}. \tag{11}$$

When $N = 1$ there are no inverse steps. Define

$$\rho = \sqrt{1 + \mu z_0^2}, \qquad x_0 = \rho^{-1}(\sqrt{\mu} z_0, 1). \tag{12}$$

This gives one objective F_μ and one unit-norm initialization for the entire run.

Lemma 2. *All roots required by* (11) *exist. They satisfy* $1/3 < z_k < \sqrt{1-2\mu} < 1$ *for* $0 \leq k \leq N-2$, *and* $0 < z_{N-1} < 1$.

Proof. The denominator of H_μ is positive. Since $\mu \le 1/200 < 1/21$,

$$H_\mu(1/3) = \frac{4-9\mu}{3+12\mu} > 1, \qquad H_\mu(\sqrt{1-2\mu}) = 0. \tag{13}$$

Every target in $(0,1)$ has a root strictly between these endpoints by continuity. The root set in the closed interval is nonempty and compact, so its largest member exists. Induction starts with $\sqrt{\mu} \in (0,1)$ and proves every inverse step. No uniqueness assertion is needed. ∎

Lemma 3. *The actual N-step orbit of F_μ from* (12) *has nonzero gradients at all its update points. Writing $v_k = x_{k,2}$, it satisfies*

$$v_k > 0, \qquad \frac{x_{k,1}}{\sqrt{\mu} v_k} = (-1)^k z_k, \qquad 0 \le k \le N-1. \tag{14}$$

Moreover,

$$v_{N-1} \ge \frac{19}{20\rho}, \qquad x_N = \frac{v_{N-1}}{4}\big((-1)^N(1-3\mu),\, 3-\mu\big). \tag{15}$$

Proof. Lemma 1 makes the actual orbit quadratic throughout. At any point with second coordinate $v > 0$, put $t = x_1/(\sqrt{\mu} v)$. Its Polyak step length and two coordinate updates are

$$\alpha(t) = \frac{1+t^2}{2(t^2+\mu)}, \tag{16}$$

$$x_{+,1} = x_1 \frac{t^2+2\mu-1}{2(t^2+\mu)}, \qquad v_+ = v\frac{(2-\mu)t^2+\mu}{2(t^2+\mu)}. \tag{17}$$

The second multiplier is positive, and the next signed ratio equals $-H_\mu(t)$. Since H_μ is odd, equations (11) and (12) prove (14) by induction. The squared gradient norm is $\mu v_k^2(z_k^2+\mu) > 0$, so every update is defined by its stated nonzero denominator.

For $k \le N-2$, Lemma 2 gives

$$\alpha(z_k) \le \tfrac{1}{2}(1+z_k^{-2}) \le 5, \qquad v_{k+1} = v_k(1-\mu\alpha(z_k)) \ge (1-5\mu)v_k. \tag{18}$$

Bernoulli's inequality therefore yields

$$v_{N-1} \ge \rho^{-1}(1-5\mu)^{N-1} \ge \rho^{-1}\big(1-5\mu(N-1)\big) \ge \frac{19}{20\rho}. \tag{19}$$

For $N = 1$ this is the empty-product estimate $v_0 = 1/\rho$, so it covers that case too. At the last input $z_{N-1}^2 = \mu$, equation (16) gives $\alpha = (1+\mu)/(4\mu)$. Substitution into (17) gives the terminal point in (15). ∎

Proof of Theorem 1. By the preceding lemmas, F_μ is admissible and the exact terminal point is still in the unit ball. Since $\rho^2 \le 1+\mu \le 201/200$ and $1-3\mu \ge 197/200$,

$$F_\mu(x_N) \ge \tfrac{1}{2}x_{N,1}^2 \ge \frac{(197/200)^2(19/20)^2}{32(201/200)} = \frac{14010049}{514560000} > \frac{1}{40}. \tag{20}$$

There is an admissible instance for every N, proving the statement about W_N. ∎

The full terminal formula also gives the identity

$$F_\mu(x_N) = \frac{(1+\mu)^3 v_{N-1}^2}{32}. \tag{21}$$

Before this final update, $v_k \le v_0 \le 1$ and $z_k \le 1$, so

$$F_\mu(x_k) = \tfrac{1}{2}\mu v_k^2(1+z_k^2) \le \mu = \frac{1}{100(N+1)}, \qquad 0 \le k < N. \tag{22}$$

Thus the construction leaves every earlier function value small and produces a large value precisely at the prescribed terminal update. It does not contradict a best-iterate estimate.

4 Comparison with the printed terminal bound

Theorem 2.3 of Zamani and Glineur [5] gives, for N projected Polyak updates on a convex set X, the bound

$$f(x^{N+1}) - f_* \le BR\,C_N, \qquad C_N = \frac{1}{\sqrt{2N+1}} \prod_{i=1}^{N} \left(\frac{4i^2}{4i^2-1}\right)^i. \tag{23}$$

Its hypotheses bound subgradient norms by B and the initial distance by R. Its step divides the exact function gap by the actual squared subgradient norm. Take $X = \mathbb{R}^2$, $B = R = 1$, and $f = F_\mu$ from Theorem 1. The projection is the identity, every subgradient is the unique gradient and is globally bounded, and the optimal value is zero. The source's x^1 and x^{N+1} are respectively our x_0 and x_N. Thus all its stated hypotheses and indices match.

The incompatibility can be made finite without evaluating a large product numerically. Since $\log(1+t) \le t$ for $t \ge 0$,

$$\log C_N \le -\tfrac{1}{2}\log(2N) + \sum_{i=1}^{N} \frac{i}{4i^2-1}. \tag{24}$$

The summands satisfy

$$\frac{i}{4i^2-1} = \frac{1}{4i} + \frac{1}{4i(4i^2-1)} \le \frac{1}{4i} + \frac{1}{12i^3}. \tag{25}$$

Using $\sum_{i=1}^{N} i^{-1} \le 1 + \log N$ and $\sum_{i=1}^{N} i^{-3} \le 1 + \int_1^\infty t^{-3}\,dt = 3/2$ gives

$$C_N \le \frac{e^{3/8}}{\sqrt{2}} N^{-1/4} \le \frac{8}{5\sqrt{2}} N^{-1/4} < \frac{6}{5} N^{-1/4}. \tag{26}$$

Here $e^{3/8} \le (1 - 3/8)^{-1} = 8/5$ follows by comparing the power series. At the explicit horizon

$$N = 64^4 = 16777216, \qquad C_N < \frac{3}{160} < \frac{1}{40} < F_\mu(x_N). \tag{27}$$

The same fixed function is used throughout this finite run, as required by the source's statement.

The proof of the cited theorem starts by imposing $\|g^k\| = B$ at every encountered point; the introduction asserts that this restriction covers the worst case. An upper bound on the norms does not imply that equality, and the counterexample rules out this reduction for the stated terminal problem. In our instance, for example, the penultimate squared gradient norm is

$$\|\nabla F_\mu(x_{N-1})\|^2 = 2\mu^2 v_{N-1}^2 < 1. \tag{28}$$

The associated step length $(1+\mu)/(4\mu)$ is large. The displayed contradiction concerns Theorem 2.3 as printed in arXiv:2407.15195v1. It makes no assertion about the validity of that paper's separate adaptive or momentum algorithms, or about its authors' distinct work on other subgradient step schedules.

5 Fixed instances and the role of the horizon

For each fixed $f \in \mathcal{C}$ and a fixed initialization, the same ordinary Polyak iteration does converge to its minimizer. Indeed $\|\nabla f\| \le 1$, so (9) implies

$$\sum_{k=0}^{\infty} f(x_k)^2 \le \|x_0\|^2. \tag{29}$$

If a zero gradient is encountered, the sequence has already reached the minimum. Otherwise (29) gives $f(x_k) \to 0$. The orbit is bounded; every cluster point is a minimizer by continuity, and the minimizer is unique. Hence $x_k \to 0$. This familiar Fejer argument is consistent with the fixed-instance discussion in [4].

Theorem 1 therefore separates fixed-instance convergence from a bound uniform over the normalized class. Its horizon parameter changes the small eigenvalue μ and the exact initial direction. The backward recursion does not restart or alter the objective during a run. Nor does it supply a single orbit whose function values fail to converge. The initialization is defined by exact algebraic roots; neither the proof nor its conclusion requires those roots to be robustly recoverable with a fixed floating-point precision.

A best-iterate estimate is also compatible with the construction. Equation (22) gives a particularly small value at every preterminal point. More generally, smooth convexity gives $\|\nabla f(x)\|^2 \le 2f(x)$, so (9) yields $\sum_{k=0}^{N-1} f(x_k) \le 2\|x_0\|^2$ and the usual $O(1/N)$ estimate for the minimum over those points. Such an estimate provides no control of the terminal value in (3).

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